

DEBDIP BANDYOPADHYAY

Senior SDET | AI Quality Engineering Lead

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PROFESSIONAL SUMMARY

Senior Software Development Engineer in Test (SDET) and AI Quality Engineering Lead with 11+ years of experience architecting enterprise test automation frameworks, CI/CD quality gates, and automated LLM/RAG evaluation systems across Enterprise ERP, Aerospace, and Generative AI platforms. Currently driving AI Evaluation and validation architecture within Cognizant's GenAI Center of Excellence (CoE), specializing in non-deterministic failure modes, synthetic adversarial evaluation, and statistical metrics (faithfulness, hallucination, answer relevance, context recall) using DeepEval and Pytest. Holds an M.Tech from IIT Jodhpur (School of AI & Data Science) with independent published research in LLM provenance, model fingerprinting, and sentence cloze scaling laws (CERN Zenodo DOIs: 10.5281/zenodo.22158286, 10.5281/zenodo.22158483).

TECHNICAL SKILLS

- AI Quality & LLM Evaluation:** LLM Evaluation Pipelines, RAG Validation, DeepEval, Semantic Similarity Assessment, Faithfulness & Hallucination Scoring, Context Relevancy, Prompt Drift, Adversarial Red-Teaming, NVIDIA NIM, PyTorch.
- Automation Frameworks:** Python, Playwright, Pytest, Pytest-BDD, Cypress, API Testing (Requests, Postman v12), Page Object Model (POM), Parallel Execution (pytest-xdist), Test Sharding, Allure Reporting.
- DevOps & Platform Operations:** Azure DevOps CI/CD Pipelines, Docker, Git, Linux, Poetry, Zero-Trust Quality Gates.
- Enterprise Systems & Methodologies:** Oracle EBS (AP, AR, PO, CE, FA), Microservices Validation, Jira/JQL, Agile/Scrum, Root Cause Analysis.

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Kolkata, India

Lead SDET / AI Evaluation Lead — GenAI Center of Excellence (CoE)

Sep 2025 – Present

- Architected end-to-end automated LLM evaluation suites using DeepEval, Pytest, and Python across production RAG and agentic workflows, replacing manual spot-checks with continuous CI/CD quality gates.
- Formalized quantitative evaluation benchmarks for non-deterministic model behaviors, establishing automated scoring pipelines across answer correctness, faithfulness, hallucination rate, and context relevancy.
- Generated 500+ adversarial synthetic test datasets to probe retrieval context boundaries, reducing pipeline hallucination escapes to staging by 35%.
- Designed zero-trust quality gating in Azure DevOps pipelines, programmatically blocking deployment builds when LLM evaluation metrics fell below defined enterprise thresholds.
- Mentored cross-functional teams of NLP practitioners, full-stack engineers, and QA automation testers on statistical evaluation strategies, synthetic data curation, and prompt boundary testing.

Senior Test Automation Engineer — Enterprise Systems / Oracle QEA

Apr 2021 – Aug 2025

- Architected an enterprise-grade test automation framework using Playwright, Python, and Pytest-BDD, establishing modular Page Object Model (POM) patterns across globally distributed enterprise platforms.
- Cut regression cycle duration by approximately 40% via execution parallelization (pytest-xdist), headless browser configuration, and strategic test sharding across Azure DevOps runners.
- Directed regression automation across 5 mission-critical Oracle EBS modules (AP, AR, PO, CE, FA) for Tier-1 global client delivery (Wyndham / American Express), ensuring consistent cross-regional ERP financial operations.
- Executed comprehensive migration test suites (migrated 183 AP and 67 PO cases; validated 500+ scripts across AP, CE, FA, PO/RI, and AR) utilizing Tricentis Tosca and UFT during high-stakes multi-region cutovers.

The Boeing Company

Bengaluru, India

Software Engineer — Test Automation

Oct 2018 – Apr 2021

- Engineered high-reliability test automation harnesses for safety-critical aerospace ground and back-office applications operating under stringent DO-178C engineering compliance standards.
- Maintained and optimized enterprise regression suites exceeding 2,000+ automated test cases, ensuring zero regression leakage across multi-phase production release cycles.
- Engineered framework modularization and test data virtualization using UFT, VBScript, and Jetty services within Microsoft TFS infrastructure.
- Honors & Awards:** Recipient of the **Boeing Pride Award (Sep 2019)** for critical automation delivery on the 737 MAX return-to-service program and the **Star Award (May 2020)** for outstanding XenDesktop beta test execution.

Accenture LLP

Kolkata / Noida / Bengaluru, India

Test Automation Engineer

Oct 2014 – Oct 2018

- Developed reusable, data-driven automation frameworks across web platforms, desktop applications, SAP MM modules, and mobile interfaces for global telecommunications and energy clients.

- Automated end-to-end integration, smoke, system regression, and User Acceptance Testing (UAT) execution cycles within tight release windows.
- Architected and launched the proprietary **One Click Capture** automated test utility, reducing third-party enterprise tooling licensing overhead by \$8,000 annually (featured in Accenture CIO internal assets, Jun 2015).
- Partnered with onshore and offshore business stakeholders to convert complex functional business requirements into high-coverage automated Proof-of-Concept (PoC) test suites.

PUBLICATIONS & AI RESEARCH PREPRINTS

- **Multi-Scale 4-Pass Sentence Cloze Infilling with Pass-Adaptive Dynamic Gating for Robust AI Text Detection**
Independent Research Preprint **CERN Zenodo DOI: 10.5281/zenodo.22158286**
Engineered a zero-shot black-box detection architecture utilizing multi-pass sentence blank-infilling and dynamic entropy gating; demonstrated statistically superior AUROC separation against dense foundation model corpora compared to standard baseline perplexity probes.
- **ClozeCongruence: Model Fingerprinting and Traceback Attribution in LLMs via Reverse Cloze Resonance**
Independent Research Preprint **CERN Zenodo DOI: 10.5281/zenodo.22158483**
Formulated an attribution framework to fingerprint generated text and determine provenance back to origin foundation model weights using reverse cloze resonance without accessing internal model activations.
- **The Cloze Scaling Laws: Prober Parameter Dynamics, Native Blank-Infilling Architecture, and Traceback**
Independent Research Preprint **SSRN Research Paper: ID 7368141**
Conducted empirical scaling-law evaluations analyzing parameter dynamics, probing token sensitivity, and boundary conditions of native blank-infilling architectures across parameter-scaled LLMs.

EDUCATION

- Master of Technology (M.Tech) in AR & VR** **Jul 2022 – Dec 2025**
Indian Institute of Technology Jodhpur (IIT Jodhpur) *Jodhpur, India*
School of Artificial Intelligence & Data Science (Executive / Industry-Sponsored Program)
Specialized coursework and advanced thesis research in deep learning representations, computer vision, and neural network evaluations.
- Bachelor of Technology (B.Tech) in Computer Science & Engineering** **Apr 2010 – Jun 2014**
Greater Noida Institute of Technology (AKTU / UPTU) *Greater Noida, India*
Core focus on data structures, algorithms, relational database systems, and software engineering methodologies.

CERTIFICATIONS & PROFESSIONAL CREDENTIALS

- **Postman API Testing Path (v12)** — Enterprise RESTful API verification, automated collection testing, and environment orchestration.
- **MIT Certified Application Tester (ASDA-MIT)** — Rigorous software quality engineering and systematic test governance.
- **Google Crash Course on Python (Coursera)** — Python programming, test automation scripting, and core data structures.